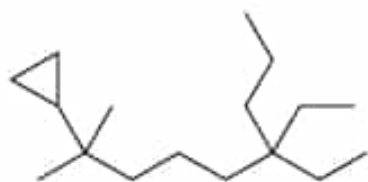


The IUPAC name of Compound **X** shown below is:



Compound **X**

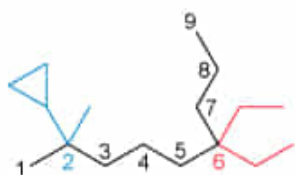
- ✓ ☒ A. 2-Cyclopropyl-6,6-diethyl-2-methylnonane [77%]
☐ B. 2-Cyclopropyl-2-methyl-6-ethylnonane [7%]
☐ C. 2-Cyclopropyl-6-ethyl-2-methyl-6-propyloctane [6%]
☐ D. 8-Cyclopropyl-4,4-diethyl-8-methylnonane [7%]

Correct

78% answered correctly

Explanation:

Alkane nomenclature



2-Cyclopropyl-6,6-diethyl-2-methylnonane

Alkanes are hydrocarbon molecules containing only single bonds and are named by identifying the parent chain (ie, the longest continuous carbon chain). If two options exist for the longest continuous chain, the parent chain is the chain that contains the greater number of substituents. The parent chain **prefix** indicates the number of carbon atoms, and its **suffix** indicates the **highest priority functional group** in the molecule. Alkanes use the suffix *-ane*.

Substituents on the parent chain are **listed alphabetically** before the parent chain using a number to indicate to which carbon on the parent chain it is bonded. The parent chain is numbered so that the substituents get the lowest number possible. If the same substituent appears more than once in a molecule, each of its numerical locations are listed in the name, and the appropriate multiplicity prefix (eg, *di-*, *tri-*) is included before the substituent name.

In this question, Compound **X** is an alkane in which the parent chain contains nine carbons (prefix *non-*) and has four substituents: a cyclopropyl, a methyl, and two ethyl groups. The parent chain is numbered from the end closest to the cyclopropyl and methyl substituents; carbon 2 has a cyclopropyl and a methyl substituent, and carbon 6 has two ethyl substituents. Therefore, the IUPAC name of Compound **X** is **2-cyclopropyl-6,6-diethyl-2-methylnonane**.

(Choice B) 2-Cyclopropyl-2-methyl-6-ethylnonane does *not* list the substituents in alphabetical order and incorrectly omits one of the ethyl substituents on carbon 6. Compound **X** contains *two* ethyl substituents, and the location of *each* substituent must be listed as well as the multiplicity prefix (in this case, *di-*) to indicate how many of that substituent are present in the molecule.

(Choice C) 2-Cyclopropyl-6-ethyl-2-methyl-6-propyloctane **incorrectly identifies** the parent chain.

(Choice D) 8-Cyclopropyl-4,4-diethyl-8-methylnonane **incorrectly numbers** the parent chain so the substituents do not have the lowest possible location positions.

Educational objective:

Alkanes are hydrocarbons containing only single bonds and are named using the suffix *-ane*. The prefix of the parent chain indicates the number of carbons in the parent chain. Substituents are listed in alphabetical order preceding the parent chain name.

Time spent:0 sec

Subject:

Organic Chemistry

QID:404156

System:

Introduction to Organic Chemistry